

Scenario-Conditional Benchmarking of Dynamic Frequency Estimators for Low-Inertia IBR Grids

Jorge Luis Mayorga Taborda, Yessica Africano Rodriguez, and Fernando Jimenez

Abstract—This paper presents a comparative benchmark of dynamic frequency estimators for low-inertia inverter-based-resource (IBR) grids under both standards-style tests and composite inverter-driven disturbances. The study covers 34 scenarios, 16 estimators, 544 tuned estimator-scenario configurations, and 60 realizations per configuration. Each estimator-scenario pair is tuned separately and then evaluated under a common dual-rate signal chain in which physics-rate waveforms are synthesized at 1 MHz and consumed by the estimators on decimated 10 kHz inputs. The paper reports six headline metrics spanning error, relay-oriented trip-risk, recovery, and computational cost, and it aggregates results by disturbance family so that ramp tracking, amplitude steps, modulation, phase discontinuities, spectral interference, ringdown, and multi-event stress remain distinguishable in the final interpretation. The main result is not a single leaderboard but a disturbance-dependent latency-robustness-risk map: SOGI-PLL leads IEEE ramps, EKF leads magnitude steps and out-of-band interference, RA-EKF leads modulation, SOGI-FLL leads phase jumps, UKF leads IBR ringdown, and PLL leads the composite IBR multi-event case on family-level RMSE. Trip-risk leaders often differ from RMSE leaders, and the hardest scenarios are the multi-event sequence and the two 60° phase-jump cases. These results show that estimator selection in low-inertia grids is governed jointly by disturbance family, protection-oriented risk, and computational budget, and that compliance-style summaries alone are insufficient for deployment decisions.

Index Terms—Frequency estimation, low-inertia power systems, inverter-based resources, Monte Carlo benchmark, relay-oriented metrics, Kalman filtering, PLL, PMU dynamics.

I. INTRODUCTION

Frequency estimation lies on the critical path of protection, disturbance monitoring, converter supervision, and fast frequency response in low-inertia grids. As inverter-based resources displace synchronous machines, the available response window after a disturbance shrinks and the measured waveform becomes less benign: the estimator may be forced to react to phase discontinuities, harmonics, out-of-band tones, oscillatory recovery, or compound inverter-driven events on the same time scale as the protection decision itself [1]–[5]. Under those conditions, an estimator that looks excellent on a nominal sinusoidal test can still accumulate large transient error, remain above relay thresholds for too long, or exceed an acceptable compute budget.

That mismatch between disturbance physics and simplified reporting is a benchmark-design problem as much as an estimator-design problem. Classical standards-style scenarios

remain necessary because they define common test points, but they do not by themselves answer the engineering question that matters in deployment: which estimator should be trusted when the disturbance family, the protection objective, and the computational budget are all explicit. A flat leaderboard cannot answer that question because it collapses cases in which the dominant failure mode is latency, cases in which it is spectral pollution, and cases in which it is transient false-trip risk.

This paper addresses that gap through a disturbance-aware benchmark study. The benchmark evaluates 34 scenarios and 16 estimators, producing 544 tuned estimator-scenario configurations and 60 Monte Carlo realizations per configuration. The study is not proposing a new estimator. It is establishing a more defensible way to compare existing estimators under low-inertia IBR stress by keeping the signal family, metric contract, and deployment objective visible from per-scenario evaluation to final interpretation.

The paper makes four concrete contributions.

- 1) It defines a benchmark protocol with per-pair tuning, dual-rate waveform generation and acquisition, scenario-wise Monte Carlo aggregation, and family-wise reporting that prevents easy families from washing out harder ones.
- 2) It organizes the study through explicit estimator and scenario taxonomies, making clear which algorithmic assumptions are being tested against ramps, magnitude steps, phase jumps, spectral interference, noisy ringdown, and composite IBR stress.
- 3) It formalizes six headline metrics spanning accuracy, worst-case transient behavior, relay-oriented trip-risk, recovery, and computational cost, and it complements descriptive comparisons with family-level statistical evidence.
- 4) It converts the benchmark outcome into deployment guidance by separating protection-oriented, low-cost synchronization-oriented, and metering-oriented operating regions rather than claiming a universal best estimator.

The central scientific claim is therefore sharper than in the shorter paper: disturbance family is part of the problem definition, not a post hoc qualifier. SOGI-PLL leads ramps, EKF leads magnitude steps and out-of-band interference, RA-EKF leads modulation, SOGI-FLL leads phase jumps, UKF leads ringdown, and PLL leads the composite IBR multi-event case on RMSE. Those winners do not coincide with the trip-risk leaders, especially under phase jumps, harmonics, and compound IBR stress. The benchmark therefore exposes a disturbance-dependent latency-robustness trade space rather than a universal champion.

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The rest of the paper is organized as follows. Section II positions the study against standards-oriented and estimator-oriented literature. Section III defines the benchmark design and evaluation protocol. Section IV explains the estimator and scenario taxonomies. Section V details the metric design and statistical analysis. Sections VI and VII present and interpret the main findings. Section VIII states the scope limits of the analysis, and the appendices provide full estimator and scenario catalogs.

II. RELATED WORK

The literature relevant to this paper spans three partially connected threads: standards-oriented dynamic tests, estimator-family design papers, and low-inertia or IBR-focused disturbance studies. The gap is not a lack of candidate algorithms. The gap is a lack of comparative studies that preserve disturbance class and operating objective at the same time.

Standards and many benchmark papers still organize evidence around isolated compliance-style disturbances. IEEE C37.118.1 and IEC/IEEE 60255-118-1 emphasize ramps, magnitude steps, modulation tests, and limited phase events [6], [7]. That logic is indispensable because it defines minimum common tests, but it also privileges clean cases in which many estimators tie on protection-oriented risk. Wright *et al.* show that even simple large phase steps can produce misleading RoCoF signatures [8]. Once such ambiguities appear, pass-fail compliance is no longer an adequate description of estimator behavior.

The estimator literature is richer on algorithms than on cross-family comparison. Window-based methods such as IpDFT, TFT, Prony, and ESPRIT trade spectral selectivity against observation delay and computational burden [9], [10]. Loop-based methods remain attractive because they are lightweight and interpretable, but their transient behavior depends strongly on gain selection, bandwidth choice, and re-lock dynamics [8], [11]. Kalman-family estimators can outperform simpler methods when the state model matches the event, yet that advantage is conditional on model adequacy and noise handling [12], [13]. Reduced-order and data-driven methods introduce additional cost-accuracy trade-offs without, by themselves, resolving how the benchmark should be interpreted [14], [15].

Recent low-inertia and IBR-focused studies widen that benchmark-design problem rather than close it. Converter-dominated systems change the time scale, disturbance composition, and recovery behavior of observable frequency dynamics [1]–[3]. Hybrid, grid-forming, and interaction-rich studies emphasize precisely those non-ideal behaviors, yet they rarely compare a broad estimator registry over standard PMU tests, non-ideal waveform stressors, composite IBR transients, relay-oriented risk metrics, and computational cost inside one coherent reporting frame [4], [5]. Four needs follow directly from that literature: event-archetype scenarios rather than only isolated textbook tests, explicit non-ideal waveform stressors, joint latency-accuracy-risk-cost reporting, and comparative summaries that remain meaningful for deployment rather than only for algorithm taxonomies.

This manuscript addresses those needs by treating benchmark structure as a scientific contribution rather than as incidental experimental detail. The paper does not claim that one metric or one scenario family is sufficient. Instead, it argues that a journal-quality benchmark must preserve three objects simultaneously: disturbance family, metric definition, and deployment objective. A benchmark that compresses these into one scalar rank may be easy to read, but it removes the design decision the reader actually needs to make.

III. BENCHMARK DESIGN

The benchmark evaluates 34 scenarios and 16 estimators, producing 544 tuned estimator-scenario configurations and 32640 Monte Carlo records. Each configuration is tuned with 100 search trials and then evaluated over 60 realizations. Scenarios are synthesized as physics-rate waveforms at 1 MHz and consumed by the estimators on decimated 10 kHz outputs. CPU cost is measured through repeated `process_time()` runs and is interpreted throughout the paper as a relative software-cost indicator rather than as a guaranteed hardware latency.

Two physical facts motivate this design. First, in low-inertia grids the initial post-disturbance frequency slope scales inversely with inertia. A compact aggregate model is

$$\frac{df}{dt} = \frac{f_s}{2H} (\Delta P - D(f - f_s)), \quad (1)$$

where f_s is nominal frequency, H is aggregate inertia, D is damping, and ΔP is the active-power imbalance [1]. Smaller H compresses the time window available to protection and control. Second, the estimator does not observe frequency directly; it infers it from the measured waveform phase,

$$f(t) \approx \frac{1}{2\pi} \frac{d\theta(t)}{dt}, \quad (2)$$

which makes the estimate sensitive to phase discontinuities, harmonics, interharmonics, amplitude modulation, and measurement noise. The benchmark is therefore designed to stress both the grid-side dynamics and the signal-side inference problem.

The benchmark is organized so that scenario identity is preserved before any global summary is formed. Each estimator-scenario pair is tuned separately, evaluated separately, and only then aggregated at the family level. That ordering matters because it prevents broad but easy families from dominating the interpretation through sheer count.

A. Design Principles

Four design rules follow from the gaps identified in Section II.

- 1) Standards-style IEEE and IEC/IEEE scenarios remain in the benchmark because they define the common compliance baseline.
- 2) Composite IBR stressors are added explicitly because realistic inverter-driven events couple discontinuities, spectral pollution, oscillatory recovery, and low-inertia transients in ways that isolated standard tests do not capture.

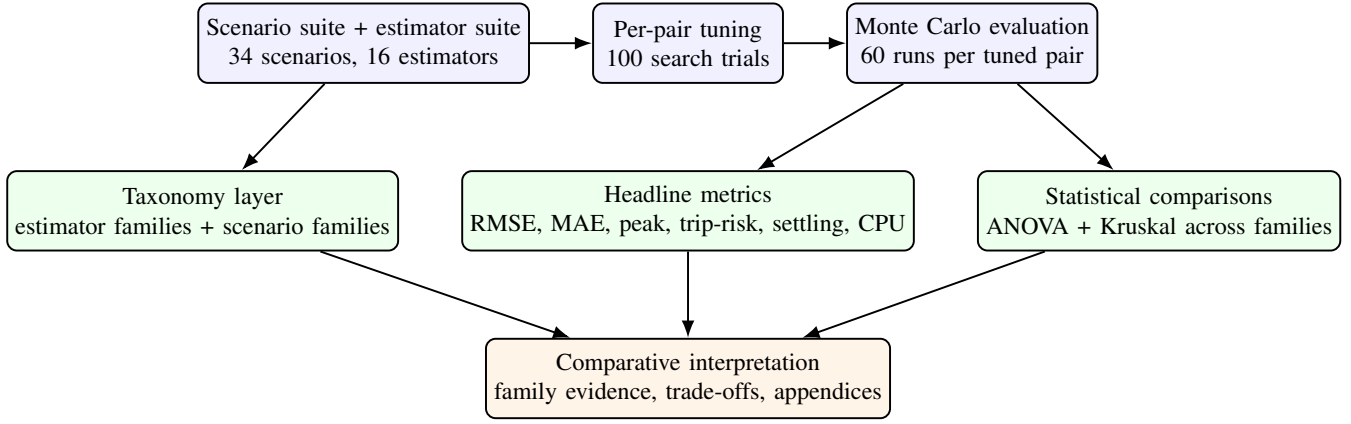


Fig. 1. Benchmark workflow used in the study. The analysis proceeds from the scenario and estimator suites through tuning, Monte Carlo evaluation, metric extraction, taxonomy-based aggregation, and statistical comparison before the final comparative interpretation is formed.

- 3) Accuracy, trip-risk, recovery, and computational cost are reported jointly because a method that is excellent on one axis can still be unusable on another.
- 4) Aggregation is performed by disturbance family before global ranking so that the benchmark answers a deployment question rather than merely producing a catalog-wide score.

One methodological point deserves emphasis. This paper is written as a benchmark study, not as a leaderboard note. The experimental design is meaningful only because disturbance family, metric definition, and deployment objective remain visible all the way from per-scenario evaluation to the final synthesis. The remainder of the paper follows that ordering explicitly.

IV. TAXONOMY AND BENCHMARK COVERAGE

The benchmark becomes readable once it is organized as two taxonomies. The first taxonomy is algorithmic: loop-based, window-based, model-based, adaptive, and data-driven estimators all occupy distinct parts of the design space. The second taxonomy is physical: clean standards-style ramps, steps, and modulation tests live beside IBR-specific harmonics, ringdowns, and composite multi-event stress. The benchmark is not large for its own sake; it is large because those axes interact.

The estimator taxonomy explains why a flat ranking is unstable. Loop-based methods are cheap and attractive for embedded deployment, but they can become fragile under large phase discontinuities. Window-based methods often look better on trip-risk or spectral pollution but pay window latency and compute. Model-based filters occupy a middle-cost regime with strong average accuracy when the state model is informative. Adaptive methods are lightweight but less robust in the hardest cases. The single data-driven method in the benchmark, Koopman (RK-DPMU), is best read as a reduced-order alternative rather than as a universal neural surrogate.

The scenario taxonomy is equally important. The clean standard families mostly separate estimators by accuracy because many methods already tie at zero trip-risk. The IBR families are different: harmonic stress, noisy ringdown, and especially

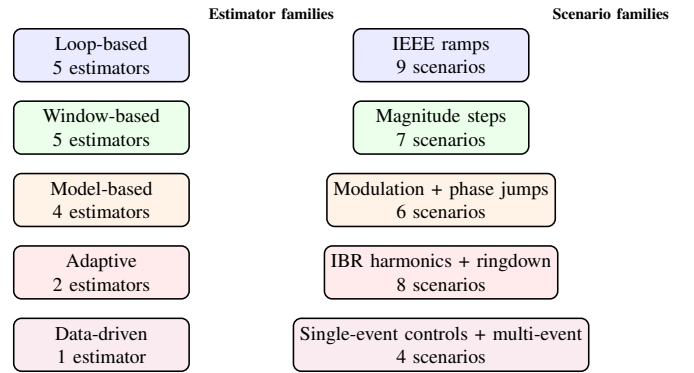


Fig. 2. Two complementary taxonomies organize the benchmark. The left side describes algorithmic assumptions; the right side describes disturbance families. Their interaction, not any single axis alone, drives the benchmark conclusions.

the composite multi-event case create disagreements between RMSE, peak error, and trip-risk leaders. That is why the full scenario catalog appears in the appendix rather than being collapsed into a one-line “dataset” description.

Figure 3 makes the reporting problem visible. A clean magnitude step and a composite IBR multi-event sequence should not be expected to produce the same ranking, even if both are called “frequency estimation” tests. The manuscript therefore treats scenario family as part of the experimental design, not as decoration around a final leaderboard. Full estimator and scenario catalogs are provided in Appendices A and B.

V. METRICS AND STATISTICAL ANALYSIS

The central methodological choice in this paper is to keep the metric contract explicit. A benchmark with heterogeneous disturbances and heterogeneous estimators cannot be summarized responsibly unless the reader can see what is being measured, how it is aggregated, and what kind of inference is being claimed.

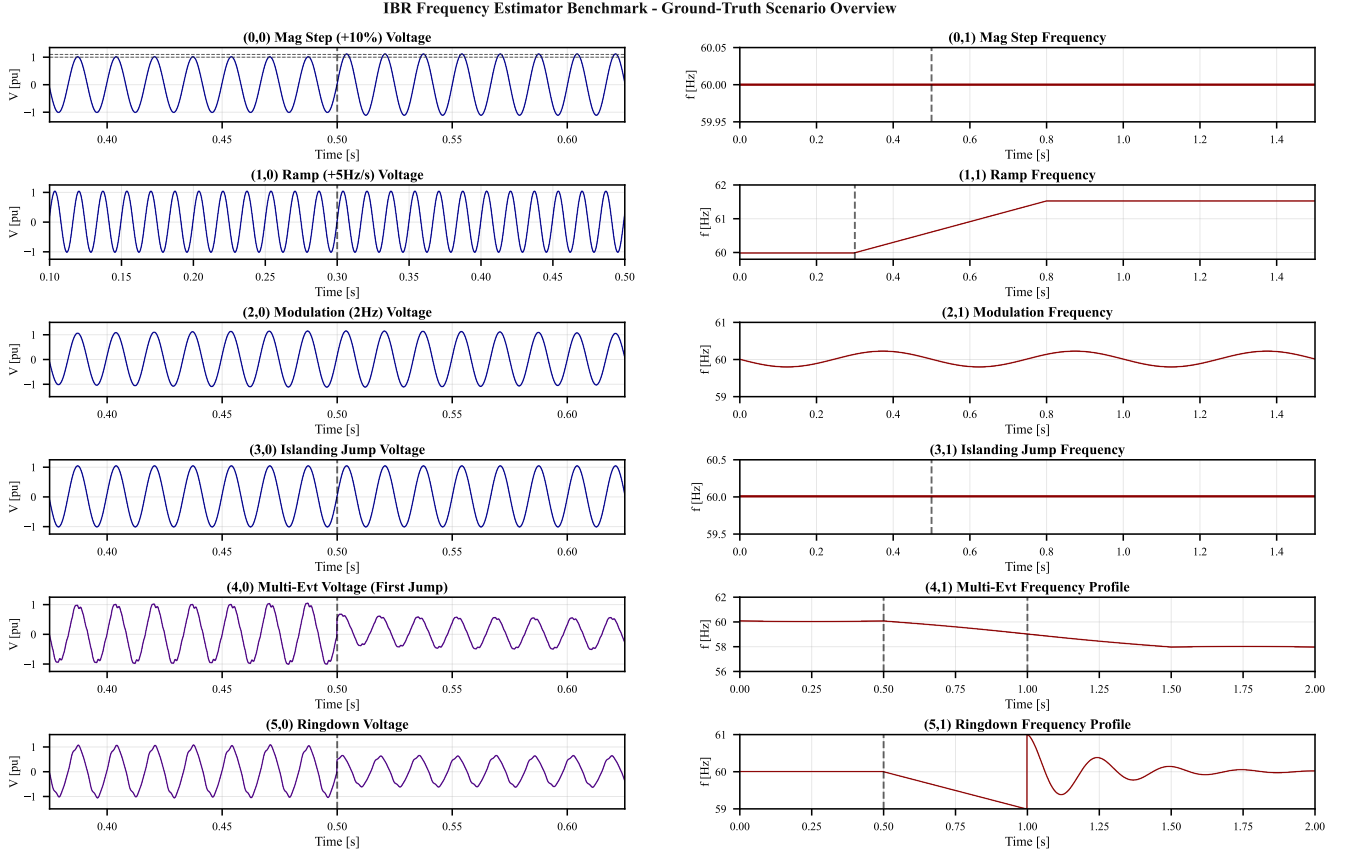


Fig. 3. Representative benchmark scenarios from the study. The figure mixes clean standard events with composite IBR stress, illustrating why the benchmark must be read by disturbance family rather than through a single aggregate score.

A. Notation and Aggregation

For scenario s , estimator j , Monte Carlo realization r , and evaluation sample $n \in \mathcal{I}_s$, let

$$e_{s,j,r}[n] = \hat{f}_{s,j,r}[n] - f_s^*[n] \quad (3)$$

denote the frequency-estimation error in hertz over the benchmark-defined evaluation window $\mathcal{I}_s$. The benchmark first computes metrics per estimator-scenario-realization tuple, then averages across Monte Carlo realizations, and only then aggregates across scenarios. That ordering matters because it preserves scenario identity before any family-level summary is formed.

For any metric m_q , the scenario-level Monte Carlo aggregate is

$$\bar{m}_{q,s,j} = \frac{1}{R} \sum_{r=1}^R m_{q,s,j,r}, \quad (4)$$

where $R = 60$ in this study. For a disturbance family $\mathcal{F}$ with scenario set $\mathcal{S}_{\mathcal{F}}$, the family summary used in the main results table is

$$M_{q,\mathcal{F},j} = \frac{1}{|\mathcal{S}_{\mathcal{F}}|} \sum_{s \in \mathcal{S}_{\mathcal{F}}} \bar{m}_{q,s,j}. \quad (5)$$

Equation (5) gives each archived scenario equal weight within its family. This prevents broad but easy families from obscuring smaller and more stressful ones through raw record count alone.

B. Headline Metrics

The benchmark tracks six headline metrics. They are intentionally heterogeneous: RMSE and MAE describe aggregate accuracy, peak error captures the worst transient excursion, trip-risk duration translates error into a relay-oriented risk quantity, settling time captures recovery behavior, and CPU time represents relative computational cost. None of these metrics dominates the others in general, and several of the benchmark's most important findings arise precisely from their disagreement.

The three basic error metrics are

$$\text{RMSE}_{s,j,r} = \sqrt{\frac{1}{|\mathcal{I}_s|} \sum_{n \in \mathcal{I}_s} e_{s,j,r}[n]^2}, \quad (6)$$

$$\text{MAE}_{s,j,r} = \frac{1}{|\mathcal{I}_s|} \sum_{n \in \mathcal{I}_s} |e_{s,j,r}[n]|, \quad (7)$$

$$\text{Peak}_{s,j,r} = \max_{n \in \mathcal{I}_s} |e_{s,j,r}[n]|. \quad (8)$$

The relay-oriented trip-risk metric is the total time for which the absolute frequency error exceeds the benchmark deadband of 0.5 Hz,

$$T_{s,j,r}^{\text{trip}} = \Delta t \sum_{n \in \mathcal{I}_s} \mathbb{I}(|e_{s,j,r}[n]| > 0.5), \quad (9)$$

where Δt is the decimated sampling interval. Settling time is defined as the first post-event time after which the esti-

mate remains inside the benchmark tolerance band for the remainder of the evaluation window. CPU time is a repeated `process_time()` measurement and should therefore be read as a relative software-cost index, not as a device-certified latency bound.

C. Family-Level Statistical Comparison

For each headline metric, the estimators are relabeled under the estimator taxonomy and tested under the null hypothesis that the estimator-family distributions are equal once the scenario summaries are pooled. We report both one-way ANOVA and Kruskal-Wallis statistics because the benchmark contains heavy-tailed and highly skewed behaviors, especially in trip-risk and severe IBR stress.

Table II shows that the estimator-family label is not cosmetic. All six null hypotheses are rejected under both ANOVA and Kruskal-Wallis tests, with Kruskal-Wallis p -values below 10^{-16} for every metric. These omnibus tests do not prove the scenario-family winner turnover that is reported later; they answer a different question. They reject the stronger null that estimator-family labels are irrelevant to the observed distributions of error, risk, settling, and cost. That result supports the paper's broader claim that flat rankings erase meaningful benchmark structure rather than merely simplifying it.

VI. RESULTS

A. No Single Estimator Dominates the Disturbance Families

Table III is the primary empirical object of the paper because it preserves disturbance family before any global compression takes place. The winner changes with the family. SOGI-PLL leads the nine IEEE ramp scenarios with a family-average RMSE of 19.4 mHz. EKF leads the seven magnitude-step scenarios with 0.388 mHz and the out-of-band interference test with 13.6 mHz. RA-EKF leads modulation at 17.8 mHz. SOGI-FLL gives the lowest family-average RMSE on phase jumps, UKF leads the IBR ringdown family, and PLL gives the lowest RMSE on the composite IBR multi-event case. Across the ten disturbance families, eight distinct estimators become RMSE winners. That turnover is too strong to be dismissed as noise around one latent global champion.

The clean standard families are also less informative than they first appear. IEEE ramps, magnitude steps, modulation, and the single-tone case contain many exact zero trip-risk ties. In those families, trip-risk acts mostly as a feasibility screen while RMSE carries the ranking burden. The IBR families behave differently. In IBR harmonics, IBR ringdown, and IBR multi-event stress, the lowest trip-risk method is often different from the lowest-RMSE method. The phase-jump family is the clearest example: SOGI-FLL leads RMSE, but ZCD yields the lowest mean trip-risk. Accuracy and protection risk are therefore coupled, but they are not interchangeable design objectives.

B. The Hardest Cases Are Composite IBR Events and Large Phase Jumps

The scenario-level difficulty ordering sharpens the family view. The hardest scenario in the study is the IBR multi-event case, with a median across-estimator RMSE of about

825.9 mHz. It is followed by the IEEE and NERC 60° phase-jump cases at about 675.1 mHz and 637.5 mHz, then by the severe IBR ringdown-noise variants, all near or above half a hertz. These cases sit orders of magnitude above the clean single-tone and magnitude-step scenarios. The benchmark is therefore spanning distinct operating regimes rather than merely repeating one test under different random seeds.

The fragmentation of winners reinforces that point. Across the 34 scenarios, ten estimators win at least one RMSE scenario outright or through exact ties. EKF is the most frequent RMSE leader with ten wins, but it is not dominant everywhere. RA-EKF and SOGI-PLL each win six scenarios, UKF and SOGI-FLL win three each, and even Koopman (RK-DPMU), ESPRIT, and Prony enter the winner set in selected cases. Scenario identity continues to matter even after family aggregation, which is exactly what a journal benchmark should surface rather than smooth away.

Figure 4 makes those relationships more explicit. The hardest-scenario panel shows that benchmark difficulty is not spread uniformly across the catalog; it is concentrated in a small set of phase-jump and IBR stress cases. The family boxplot shows the same effect structurally: clean standard families occupy the low-error region, whereas IBR ringdown and multi-event stress define the upper tail. The lower panels show why the final recommendation cannot come from one metric alone. Estimators that look attractive on RMSE versus trip-risk are not the same estimators that dominate the mean-rank versus CPU view.

The appendix extends that compact picture into a family-resolved atlas. Figures 6 and 7 keep OOB interference, single-tone, and frequency-step baselines in separate panels rather than collapsing them into a single control bucket, and they plot the five lowest-RMSE estimators within each family across the benchmark scenarios.

C. Global Summaries Help Only When Read as Secondary Evidence

Global summaries are still useful, but only as secondary evidence. On mean rank across scenarios, TFT is the best RMSE method in the study, followed by RA-EKF and EKF. That is a different story from the family winner map because mean rank rewards repeated near-front finishes rather than dominance in the hardest families. The difference is not a contradiction; it shows that consistency and absolute error severity are different quantities.

The Pareto set in Table IV reveals the same distinction from a cost perspective. EKF and RA-EKF occupy the low-error middle-cost region. SOGI-PLL and SOGI-FLL remain attractive because their compute cost is at least an order of magnitude smaller. ZCD survives on the Pareto set only because its CPU cost is nearly negligible; its global error is far too large for that Pareto membership to be read as a practical recommendation. The mathematical Pareto frontier is therefore broader than the operationally credible frontier.

Figure 5 is best read in that conditional way. The estimator that looks strongest in the ramp case is not the one that best handles the multi-event sequence or the noisy ringdown

TABLE I
HEADLINE METRICS ANALYZED IN THE BENCHMARK.

Code	Name	What it measures	Interpretation
m1_rmse_hz	RMSE	Global accuracy over the evaluation window.	Lower is better.
m2_mae_hz	MAE	Average absolute error without squaring large excursions.	Lower is better.
m3_max_peak_hz	Peak error	Worst instantaneous frequency error.	Lower is better.
m5_trip_risk_s	Trip-risk duration	Total time above the 0.5 Hz relay deadband.	Lower is better.
m8_settling_time_s	Settling time	Time until the estimate stays within the tolerance band.	Lower is better.
m13_cpu_time_us	CPU time	Average per-sample process-time cost.	Lower is better.

TABLE II
OMNIBUS FAMILY-COMPARISON TESTS COMPUTED FROM THE SCENARIO SUMMARIES AFTER RELABELING ESTIMATORS WITH THE STUDY TAXONOMY. THE NULL HYPOTHESIS FOR EVERY ROW IS THAT ESTIMATOR-FAMILY DISTRIBUTIONS ARE EQUAL FOR THE REPORTED METRIC.

Metric	Families	ANOVA F	p	Kruskal H	p
RMSE	5	12.13	1.89e-09	157.02	$< 10^{-16}$
MAE	5	16.04	2.05e-12	174.72	$< 10^{-16}$
Peak error	5	5.49	2.43e-04	125.19	$< 10^{-16}$
Trip-risk	5	89.56	$< 10^{-16}$	170.87	$< 10^{-16}$
Settling time	5	17.25	2.52e-13	92.21	$< 10^{-16}$
CPU time	5	71.31	$< 10^{-16}$	442.87	$< 10^{-16}$

TABLE III
FAMILY-LEVEL RMSE AND TRIP-RISK SUMMARY FROM THE 34-SCENARIO BENCHMARK SUITE. RMSE VALUES ARE FAMILY MEANS OF SCENARIO MEANS AND ARE REPORTED IN MHz. THE UNCERTAINTY TERM IS THE WINNER’S MEAN WITHIN-SCENARIO MONTE CARLO STANDARD DEVIATION ACROSS THE FAMILY.

Family	Count	RMSE winner	RMSE [mHz]	Runner-up	Δ [mHz]	Lowest T_{trip} [s]
IEEE ramps	9	SOGI-PLL	19.412 ± 6.367	Koopman (RK-DPMU)	2.677	0 (11-way tie)
Magnitude steps	7	EKF	0.388 ± 0.143	RA-EKF	1.451	0 (10-way tie)
Modulation	3	RA-EKF	17.781 ± 21.411	Koopman (RK-DPMU)	13.659	0 (8-way tie)
Phase jumps	3	SOGI-FLL	174.052 ± 67.466	PLL	4.031	ZCD: 0.0194
OOB interference	1	EKF	13.641 ± 11.132	RA-EKF	7.352	0 (2-way tie)
Single-tone	1	ESPRIT	0 ± 0	Koopman (RK-DPMU)	0	0 (13-way tie)
IBR harmonics	3	EKF	70.464 ± 27.297	TFT	4.126	TFT: 0.0095
IBR ringdown	5	UKF	313.127 ± 23.247	EKF	6.667	ESPRIT: 0.0989
IBR multi-event	1	PLL	173.495 ± 66.708	UKF	2.737	IPDFT: 0.0439
Frequency step	1	Koopman (RK-DPMU)	89.865 ± 58.051	LKF2	5.291	Koopman (RK-DPMU): 0.0099

TABLE IV
GLOBAL RMSE-VERSUS-CPU PARETO SET EXTRACTED FROM THE BENCHMARK.

Estimator	Family	RMSE [mHz]	CPU [μ s]
RA-EKF	Model-based	116.19	1.610
EKF	Model-based	123.20	0.813
SOGI-PLL	Loop-based	188.94	0.087
SOGI-FLL	Loop-based	307.87	0.035
ZCD	Loop-based	1.69e+05	0.017

family, and the estimator that minimizes trip-risk is not always the one that minimizes RMSE. Once those panels are placed beside the family table and the omnibus tests, the benchmark’s main claim becomes difficult to dismiss: scenario family and operating objective jointly determine what “best” means.

VII. DISCUSSION

The main value of the benchmark is not that it names a winner; it constrains the decision space. Read in that way, the results support three operational regions rather than one champion list. First, EKF, RA-EKF, and UKF define the protection-

oriented model-based region. They are not the cheapest estimators, but they remain the most credible broad-coverage choice when the disturbance class is uncertain and transient recovery matters as much as steady-state error. Second, SOGI-PLL, SOGI-FLL, and PLL define the low-cost synchronization region. They remain compelling when embedded implementation, deterministic execution, or ramp tracking dominates the design brief, but their behavior deteriorates more sharply under large discontinuities and composite IBR stress. Third, window-based methods such as TFT, IPDFT, Prony, and ESPRIT define the metering-oriented region: they can offer attractive steady-state precision or spectral behavior, but they pay for that strength through observation delay and, in several families, weaker protection-facing transient behavior.

Three practical decision rules follow. First, when the disturbance family is known in advance, the family-level table is the correct entry point because it preserves the relevant operating regime. Second, when the disturbance family is uncertain but the compute budget is moderate, the model-based filters are the safest broad-coverage choice in this study. Third, when the implementation budget is severe, low-cost

Additional plots generated directly from the archival benchmark JSON

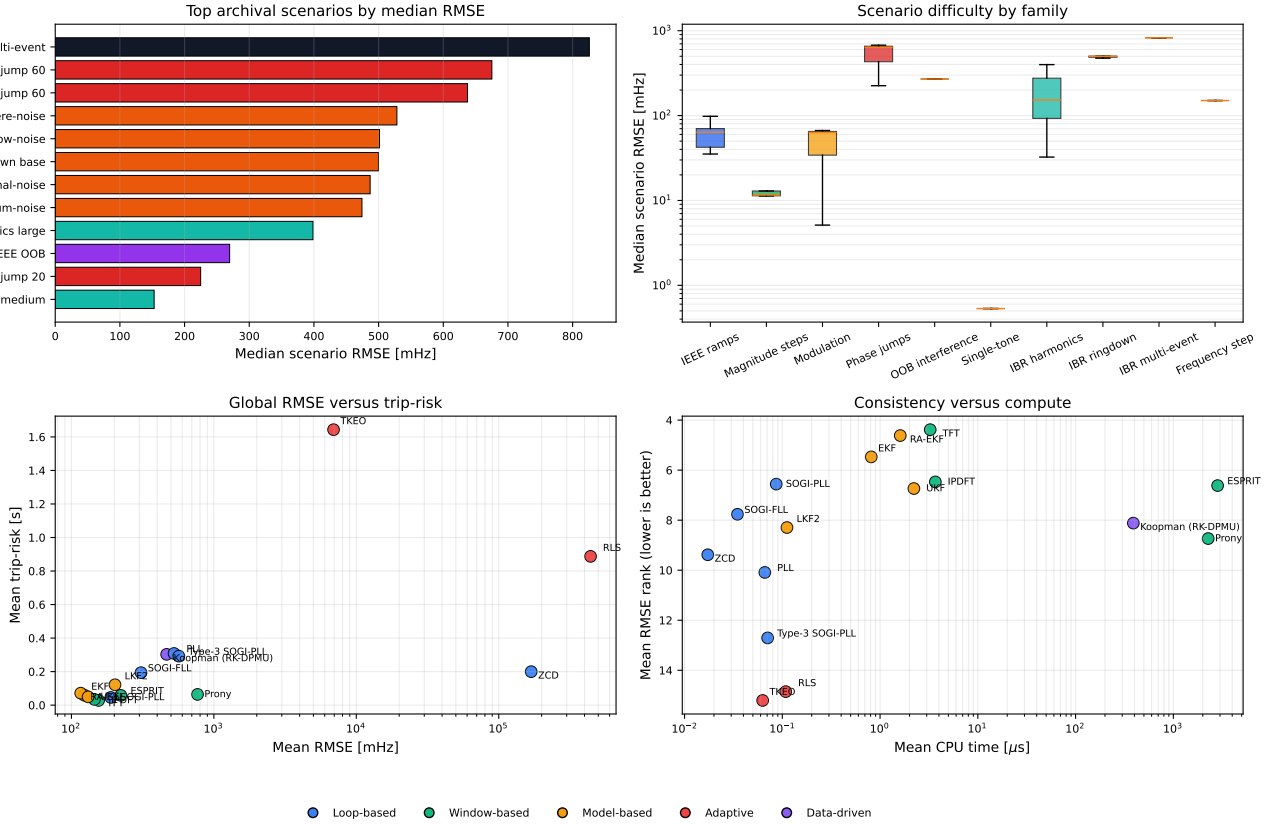


Fig. 4. Additional benchmark views. Top left: the hardest scenarios concentrate in composite IBR stress and large phase jumps. Top right: scenario-family difficulty spans more than three orders of magnitude. Bottom left: low RMSE and low trip-risk do not identify the same estimators. Bottom right: mean-rank consistency and compute cost define separate deployment regions.

loop methods remain credible only if the application can tolerate their weaker behavior under large discontinuities and composite IBR stress. These rules are more useful than a single leaderboard because they map the benchmark back to concrete protection, synchronization, and metering choices.

The statistical comparison layer strengthens that interpretation, but it should be read with care. The omnibus tests do not certify individual winners, and they do not replace the family-level tables. Their role is narrower and still important: they reject the idea that estimator-family labels are incidental metadata once the scenario summaries are pooled. That is enough to justify a reporting frame that honors family structure instead of flattening it.

One subtle implication concerns the role of headline winners. PLL leads the composite IBR multi-event family on RMSE, yet it is not the most robust global method. Koopman leads the isolated frequency-step case, yet that result does not overturn the broader pattern. ZCD enters the Pareto set, yet it does so for a reason that is mathematically true and operationally weak: vanishing CPU time paired with very poor error. A careful journal benchmark must distinguish these statements. Winning a local subproblem, surviving a Pareto screen, and offering a robust deployment recommendation are not equivalent achievements.

The older standards-style families also need to be interpreted in context. Clean ramps and magnitude steps remain necessary because they expose structural latency and tuning behavior under controlled conditions. However, they are not sufficient because they understate the failure modes that dominate low-inertia IBR operation: large phase discontinuities, noisy ringdown, and multi-event stress. The benchmark therefore supports a layered validation logic: standards-style cases establish baseline conformance, while IBR-specific and composite cases determine whether an estimator remains trustworthy when the signal assumptions behind that conformance are violated.

VIII. LIMITATIONS

The benchmark is broad, but it remains finite. Its conclusions apply to the 34-scenario suite and the 16 estimators studied here; they should not be read as universal claims over every future estimator variant or every possible disturbance class.

The benchmark is also still a software benchmark. CPU time is relative desktop cost, not hardware-in-the-loop latency. The family-level statistical analysis is omnibus rather than causal or pairwise-complete. Real-time deployment claims therefore remain conditional, and a future hardware campaign would

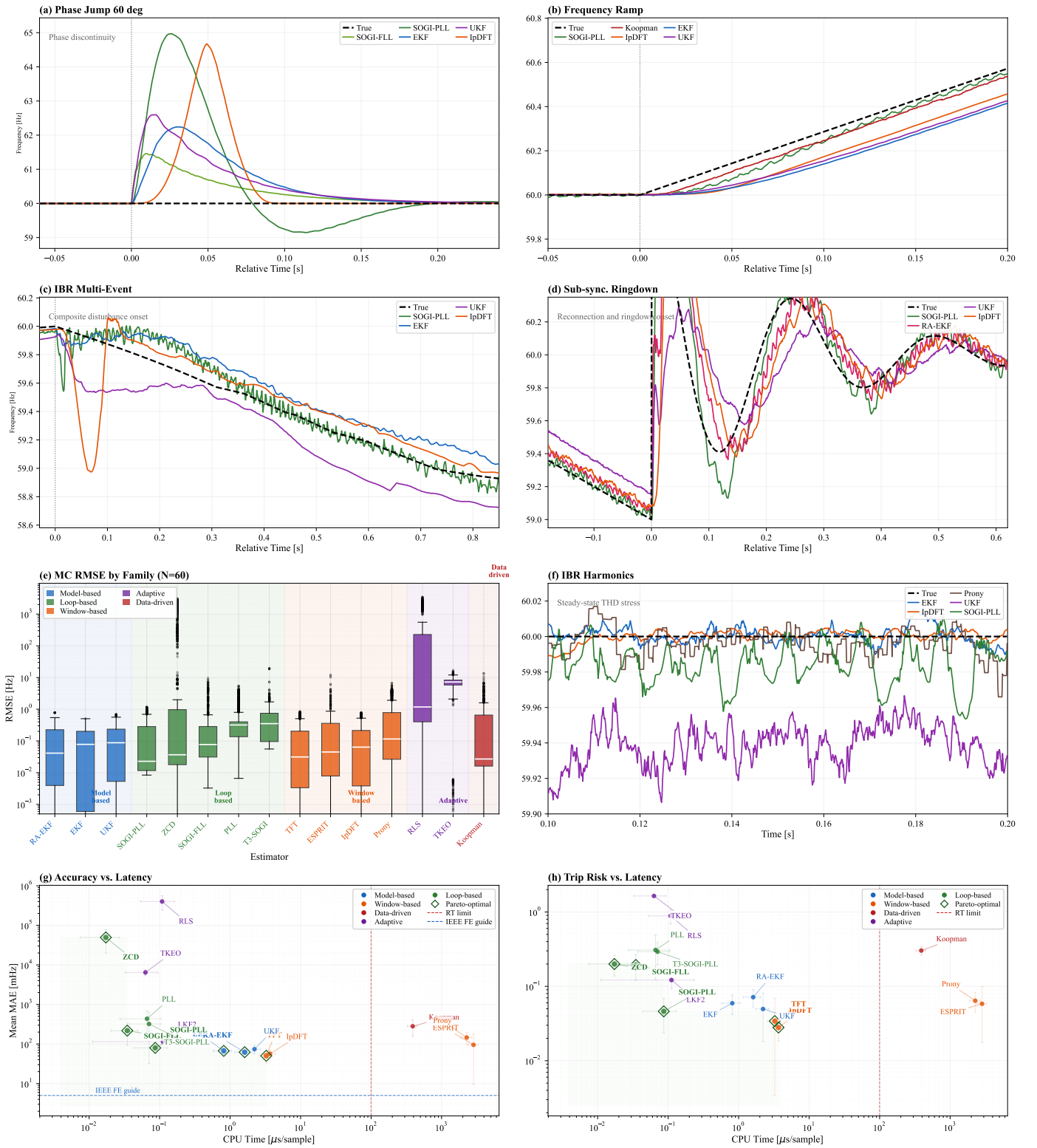


Fig. 5. Representative views from the benchmark. The time-domain panels show why scenario family matters, while the global panels expose trade-offs between accuracy, risk, and cost that cannot be captured by a single scalar score.

be needed before translating the present ordering directly into device-level guarantees.

Finally, the benchmark is strongest on the interaction between standard PMU-style disturbances and IBR-oriented harmonic, ringdown, and composite stress. It says less about

multi-phase faults, networked estimator fusion, or plant-specific controller nonlinearities. Those are natural extension points rather than defects, but they should be stated openly.

IX. CONCLUSION

This paper establishes a disturbance-aware benchmark reading of dynamic frequency estimation for low-inertia IBR grids. The central result is clear: no single estimator dominates once the benchmark spans clean standards-style tests, IBR harmonics, noisy ringdown, and composite multi-event stress. SOGI-PLL, EKF, RA-EKF, SOGI-FLL, UKF, PLL, ESPRIT, and Koopman (RK-DPMU) each lead at least one disturbance family on RMSE, while the trip-risk leaders often differ from the RMSE leaders. The benchmark therefore resolves not to one global winner, but to a structured map of conditional strengths and weaknesses.

Three conclusions follow. First, a latency-robustness trade-off remains structural rather than accidental: methods that suppress spectral pollution through longer memory or heavier filtering often pay for it through slower transient recovery. Second, compliance-style scenarios are necessary but insufficient. The large phase-jump and composite IBR cases expose failure modes that are largely invisible in cleaner standard families. Third, deployment recommendations must remain objective-aware. If the design priority is low-cost ramp tracking, loop methods remain competitive. If the estimator must remain broadly trustworthy under mixed disturbances, the model-based filters are the strongest default recommendation in this study. If the priority is spectral precision or post-event analysis, some window-based methods remain attractive despite their higher delay.

The methodological consequence is equally important. Benchmark conclusions should report the disturbance family, metric contract, and deployment objective explicitly; otherwise, a single ranking can hide the very behavior the benchmark was designed to expose. That disturbance-aware, objective-aware reading is the main contribution of the study.

APPENDIX A
ESTIMATOR CATALOG

Table V provides a compact explanation of every estimator included in the benchmark. The point of the table is not to claim that one-line descriptions replace a full implementation description, but to keep the manuscript self-contained: a reader should be able to understand what each method assumes, where it helps, and what trade-off it carries without leaving the paper.

TABLE V: Estimator catalog for the benchmark. RMSE wins count exact scenario-level wins in the 34-scenario suite.

Estimator	Family	Principle	Where it helps	Main trade-off	RMSE wins
EKF	Model-based	Extended Kalman filter for single-phase dynamic frequency tracking.	Most RMSE wins and strongest overall balance of error and cost.	Model mismatch can still hurt under severe composite stress.	10
ESPRIT	Window-based	Subspace rotational-invariance spectral estimator.	Frequent trip-risk leader in difficult scenarios.	By far the most expensive routine in the benchmark.	1
IPDFT	Window-based	Interpolated DFT around the dominant spectral bin.	Good steady-state selectivity and low trip-risk in several hard cases.	Observation-window latency and transient smearing.	0
Koopman (RK-DPMU)	Data-driven	Reduced-order data-driven estimator inspired by robust Koopman models.	Occasional scenario wins and competitive ramp behavior.	High compute without broad global RMSE dominance.	2
LKF2	Model-based	Linear Kalman filtering with a reduced sinusoidal state model.	Low compute for a state-space estimator.	Less robust than EKF/UKF/RA-EKF on the hardest nonlinear events.	0
PLL	Loop-based	Single-phase phase-locked loop tracking instantaneous phase.	Good composite-event RMSE and low embedded cost.	Can slip or overshoot under large phase discontinuities.	2
Prony	Window-based	Parametric modal fit from short signal windows.	Useful harmonic and trip-risk behavior in selected cases.	Heavy compute and fragile numerical conditioning.	1
RA-EKF	Model-based	RoCoF-augmented and robustness-tuned extended Kalman filter.	Best modulation-family RMSE and strong global trade-off profile.	Higher cost than the cheapest loop methods.	6
RLS	Adaptive	Recursive least-squares fit of an AR(2) sinusoidal model.	Online adaptation with low nominal compute.	Unstable under the hardest composite disturbances in this study.	0
SOGI-FLL	Loop-based	SOGI front-end with frequency-locked loop update law.	Very low compute and strong phase-jump behavior.	Less competitive than the best model-based filters on clean families.	3
SOGI-PLL	Loop-based	Second-order generalized integrator front-end plus PLL.	Best ramp-family RMSE with sub-microsecond cost.	Not universally best under IBR ringdown or multi-event stress.	6
TFT	Window-based	Taylor-Fourier expansion for dynamic phasor estimation.	Strong global mean-rank behavior across error metrics.	Higher compute than loop methods and window dependence.	2

Estimator	Family	Principle	Where it helps	Main trade-off	RMSE wins
TKEO	Adaptive	Teager-Kaiser energy operator for instantaneous frequency cues.	Fast lightweight adaptive baseline.	Limited robustness across difficult benchmark families.	0
Type-3 SOGI-PLL	Loop-based	Type-3 SOGI-PLL variant tuned for low-inertia operation.	Cheap embedded implementation with improved loop shaping.	Rarely reaches the front of the benchmark on accuracy.	0
UKF	Model-based	Unscented Kalman filter with nonlinear sigma-point propagation.	Best RMSE in the ringdown-noise family.	More compute than EKF with no universal advantage.	3
ZCD	Loop-based	Zero-crossing frequency readout from waveform sign changes.	Minimum compute and transparent implementation.	Sensitive to discontinuities, noise, and harmonics.	0

APPENDIX B SCENARIO CATALOG

Table VI lists every scenario contained in the benchmark suite. The generic IEEE ramp and magnitude-step cases are kept alongside their parameterized variants because the study treats them as separate scenario IDs.

TABLE VI: Scenario catalog for the benchmark suite. Difficulty is the median scenario RMSE across estimators in mHz.

Scenario ID	Family	Benchmark purpose	Median RMSE [mHz]
IEEE_Freq_Ramp	IEEE ramps	Default IEEE frequency ramp benchmark.	66.15
IEEE_Freq_Ramp_0.25Hzs	IEEE ramps	0.25 Hz/s sustained RoCoF.	42.34
IEEE_Freq_Ramp_0.5Hzs	IEEE ramps	0.5 Hz/s sustained RoCoF.	35.24
IEEE_Freq_Ramp_10Hzs	IEEE ramps	10 Hz/s sustained RoCoF.	98.02
IEEE_Freq_Ramp_15Hzs	IEEE ramps	15 Hz/s sustained RoCoF.	60.69
IEEE_Freq_Ramp_1Hzs	IEEE ramps	1 Hz/s sustained RoCoF.	39.56
IEEE_Freq_Ramp_20Hzs	IEEE ramps	20 Hz/s sustained RoCoF.	70.31
IEEE_Freq_Ramp_2Hzs	IEEE ramps	2 Hz/s sustained RoCoF.	63.67
IEEE_Freq_Ramp_5Hzs	IEEE ramps	5 Hz/s sustained RoCoF.	81.90
IEEE_Mag_Step	Magnitude steps	Default IEEE magnitude-step benchmark.	12.93
IEEE_Mag_Step_10pct	Magnitude steps	10 percent magnitude change.	11.59
IEEE_Mag_Step_15pct	Magnitude steps	15 percent magnitude change.	12.93
IEEE_Mag_Step_1pct	Magnitude steps	1 percent magnitude change.	11.29
IEEE_Mag_Step_25pct	Magnitude steps	25 percent magnitude change.	11.40
IEEE_Mag_Step_50pct	Magnitude steps	50 percent magnitude change.	11.29

Scenario ID	Family	Benchmark purpose	Median RMSE [mHz]
IEEE_Mag_Step _5pct	Magnitude steps	5 percent magnitude change.	12.93
IEEE_Modulation	Modulation	Combined amplitude and phase modulation benchmark for measurement bandwidth.	63.26
IEEE_Modulation _AM	Modulation	Pure amplitude modulation benchmark.	5.13
IEEE_Modulation _FM	Modulation	Pure phase/frequency modulation benchmark.	66.67
IEEE_Phase_Jump _20	Phase jumps	Moderate 20 degree phase discontinuity aligned with IEEE anti-islanding limits.	224.87
IEEE_Phase_Jump _60	Phase jumps	Large 60 degree phase discontinuity stress test.	675.14
NERC_Phase_Jump _60	Phase jumps	Extreme 60 degree islanding jump with additional harmonic stress.	637.49
IEEE_OOB _Interference	OOB interference	Out-of-band interference rejection test.	269.66
IEEE_Single _SinWave	Single-tone	Clean steady-state sinusoid used as a deterministic baseline.	0.53
IBR_Harmonics _Large	IBR harmonics	Severe harmonic pollution representative of stressed IBR operation.	398.59
IBR_Harmonics _Medium	IBR harmonics	Medium harmonic content representative of realistic grid-edge distortion.	152.96
IBR_Harmonics _Small	IBR harmonics	Small harmonic pollution representative of IEEE 519 compliant conditions.	32.44
IBR_Power _Imbalance _Ringdown	IBR ringdown	Nominal IBR power-imbalance ringdown with oscillatory recovery.	499.75
IBR_Power _Imbalance _Ringdown_Low _Noise	IBR ringdown	Low-noise ringdown variant with mild interharmonic contamination.	501.38
IBR_Power _Imbalance _Ringdown _Medium_Noise	IBR ringdown	Medium-noise ringdown variant designed to stress loop robustness.	474.29
IBR_Power _Imbalance _Ringdown _Normal_Noise	IBR ringdown	Normal-noise ringdown variant close to the default operating point.	486.90
IBR_Power _Imbalance _Ringdown _Severe_Noise	IBR ringdown	Severe-noise ringdown variant with the strongest interharmonic stress.	528.32
IBR_Multi_Event	IBR multi-event	Composite IBR stress sequence combining phase jump, ramp, harmonics, and oscillatory recovery.	825.86
IEEE_Freq_Step	Frequency step	IEC/IEEE frequency step benchmark for abrupt nominal-frequency change.	150.43

APPENDIX C FAMILY RMSE ATLAS

Figures 6 and 7 provide the family-resolved view requested by the benchmark story in the main text. Each family keeps its own scenario axis, the three small baseline families are not merged, and every panel shows the five best RMSE estimators inside that family rather than a global top-five list.

Family-resolved RMSE atlas from the archival benchmark JSON (page 1/2)

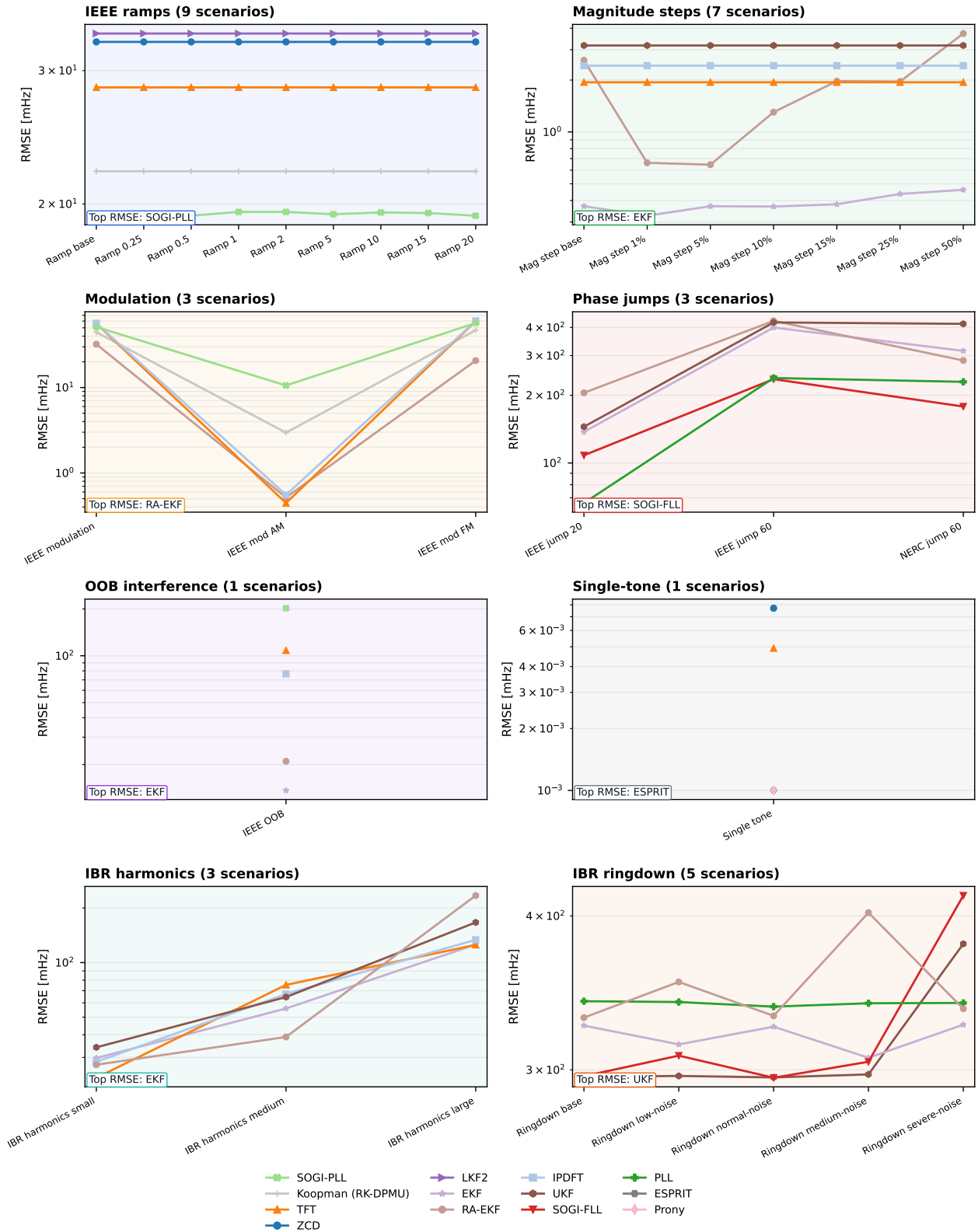


Fig. 6. Family-resolved RMSE atlas from the benchmark suite (page 1/2). Each panel keeps one scenario family separate, including the small baseline families, and plots the five lowest-RMSE estimators for that family across its scenarios. The vertical axis is log-scaled in mHz. Families on this page: IEEE ramps, Magnitude steps, Modulation, Phase jumps, OOB interference, Single-tone, IBR harmonics, IBR ringdown.

Family-resolved RMSE atlas from the archival benchmark JSON (page 2/2)

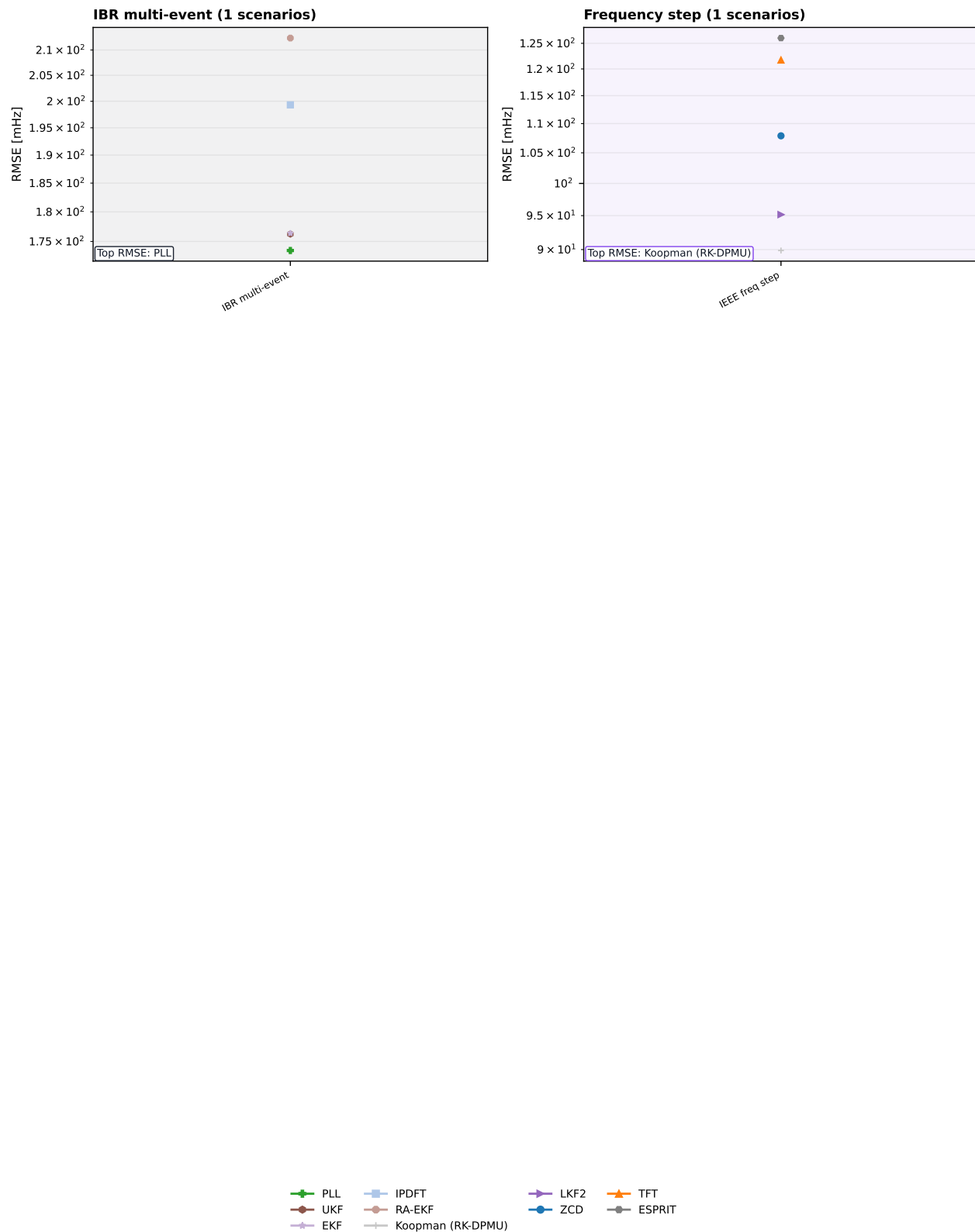


Fig. 7. Family-resolved RMSE atlas from the benchmark suite (page 2/2). Each panel keeps one scenario family separate, including the small baseline families, and plots the five lowest-RMSE estimators for that family across its scenarios. The vertical axis is log-scaled in mHz. Families on this page: IBR multi-event, Frequency step.

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